mandate

Scoped USDC payments for AI agents



2026 FinTechathon / Topic F

Mandate Labs

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Spending authority for AI agents

The owner defines the task

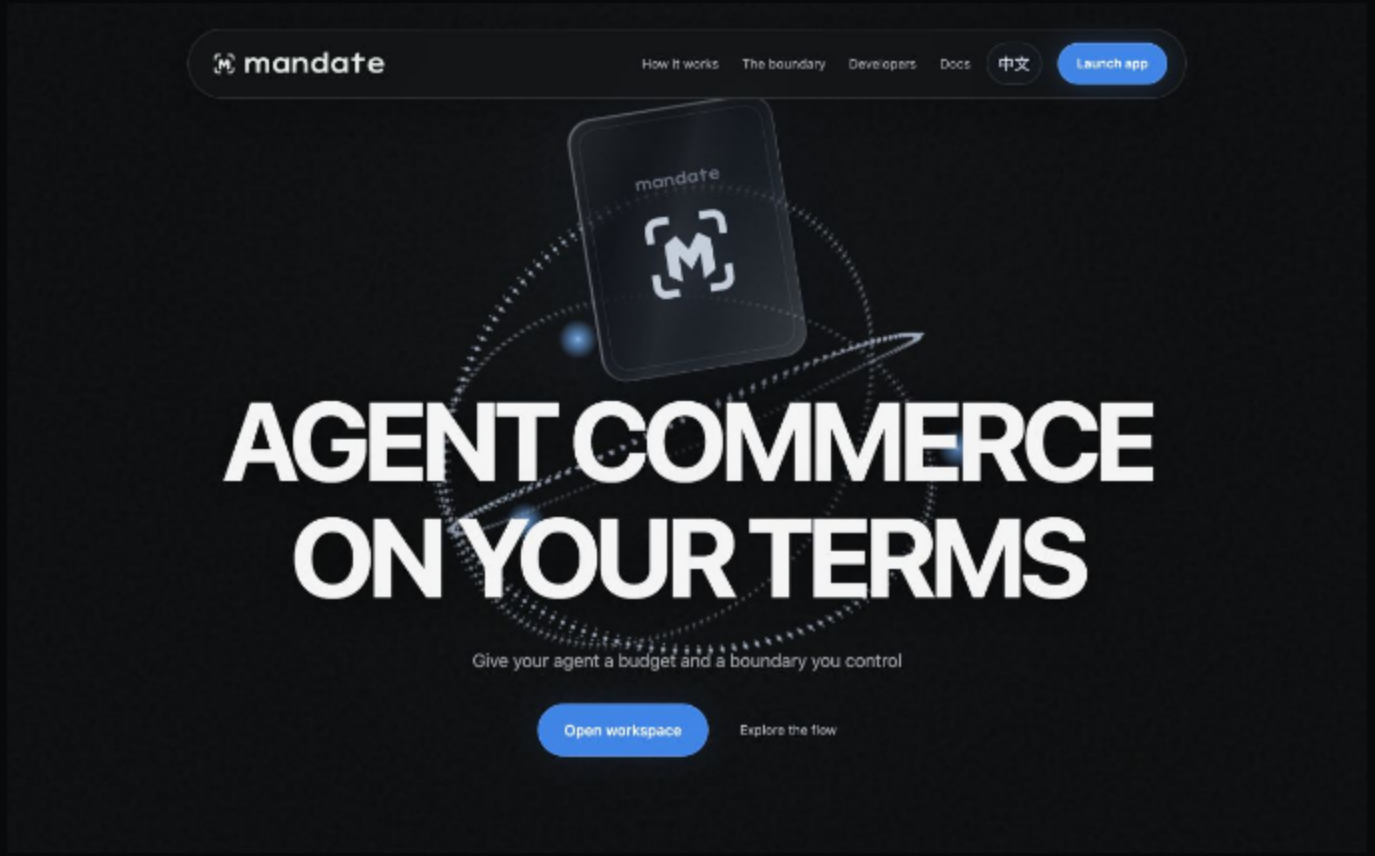
Approve services, recipient, asset, budget and expiry.

The agent executes

Use a limited session to purchase approved steps.

The contract enforces the financial boundary during execution.

The workspace and Claude share one task



Owner workspace

Review permissions
and payment outcomes

Claude Desktop MCP

Run the approved task
through a local gateway

Owner keys remain
in the external wallet

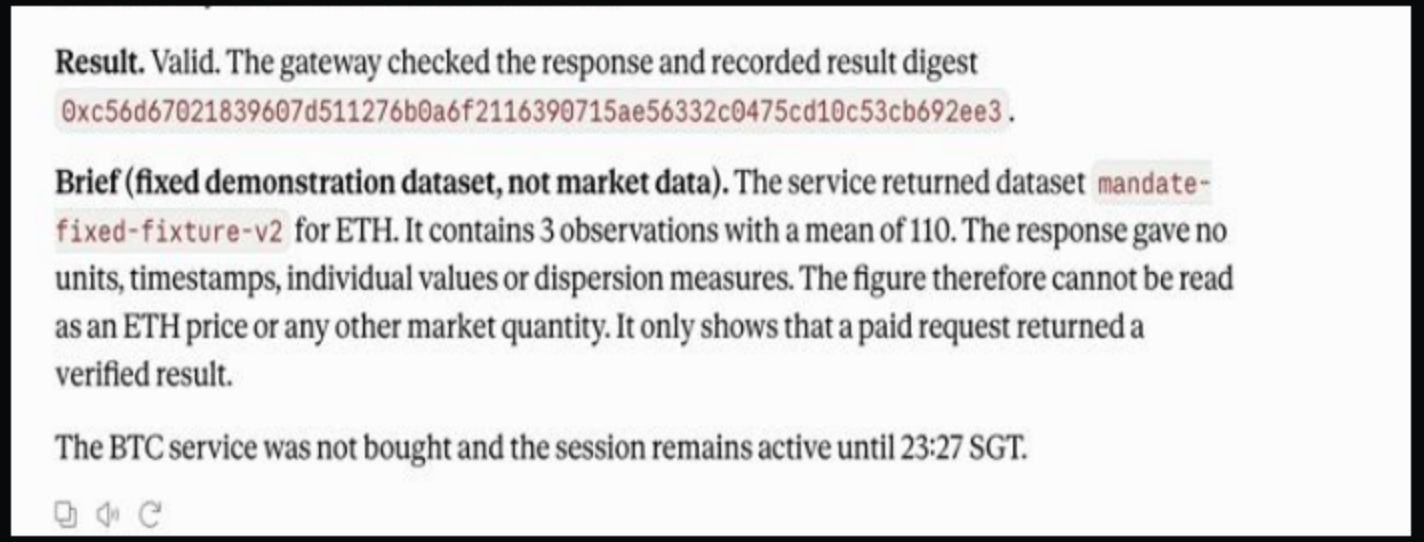
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Claude purchases an ETH statistics sample

0.15
test USDC budget

0.10
per-payment maximum

0.01
ETH sample price



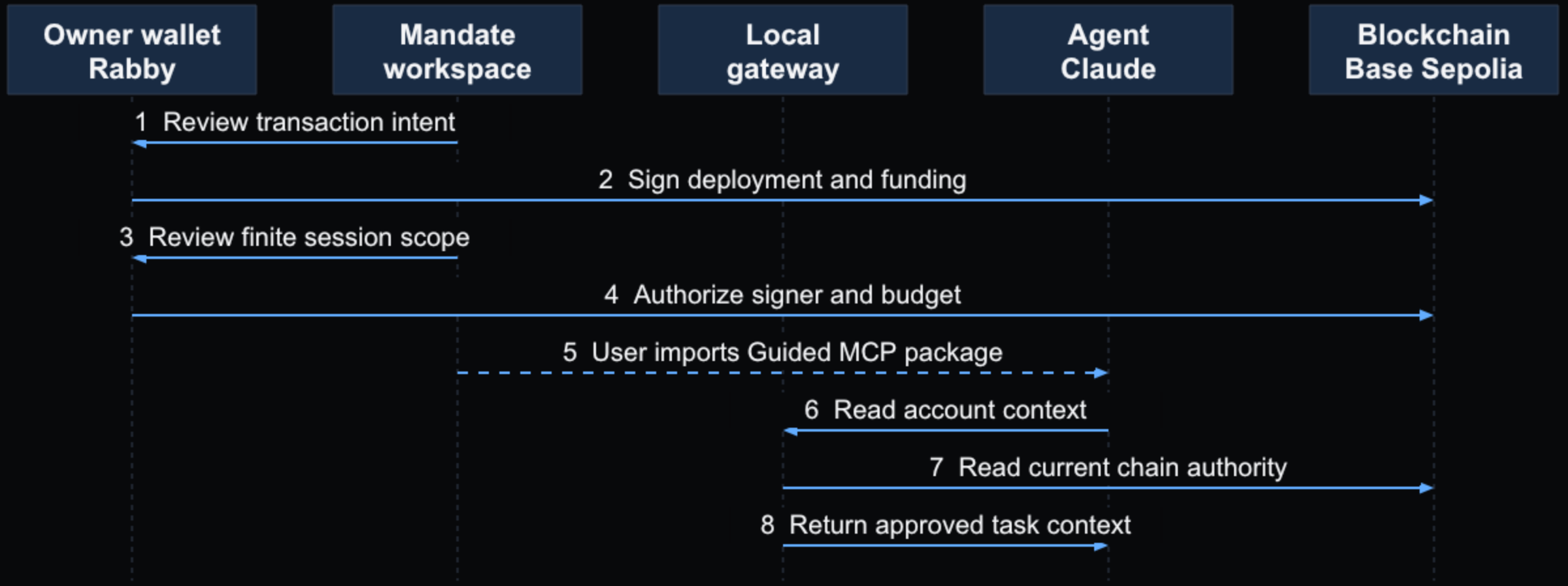
Result. Valid. The gateway checked the response and recorded result digest `0xc56d67021839607d511276b0a6f2116390715ae56332c0475cd10c53cb692ee3`.

Brief (fixed demonstration dataset, not market data). The service returned dataset `mandate-fixed-fixture-v2` for ETH. It contains 3 observations with a mean of 110. The response gave no units, timestamps, individual values or dispersion measures. The figure therefore cannot be read as an ETH price or any other market quantity. It only shows that a paid request returned a verified result.

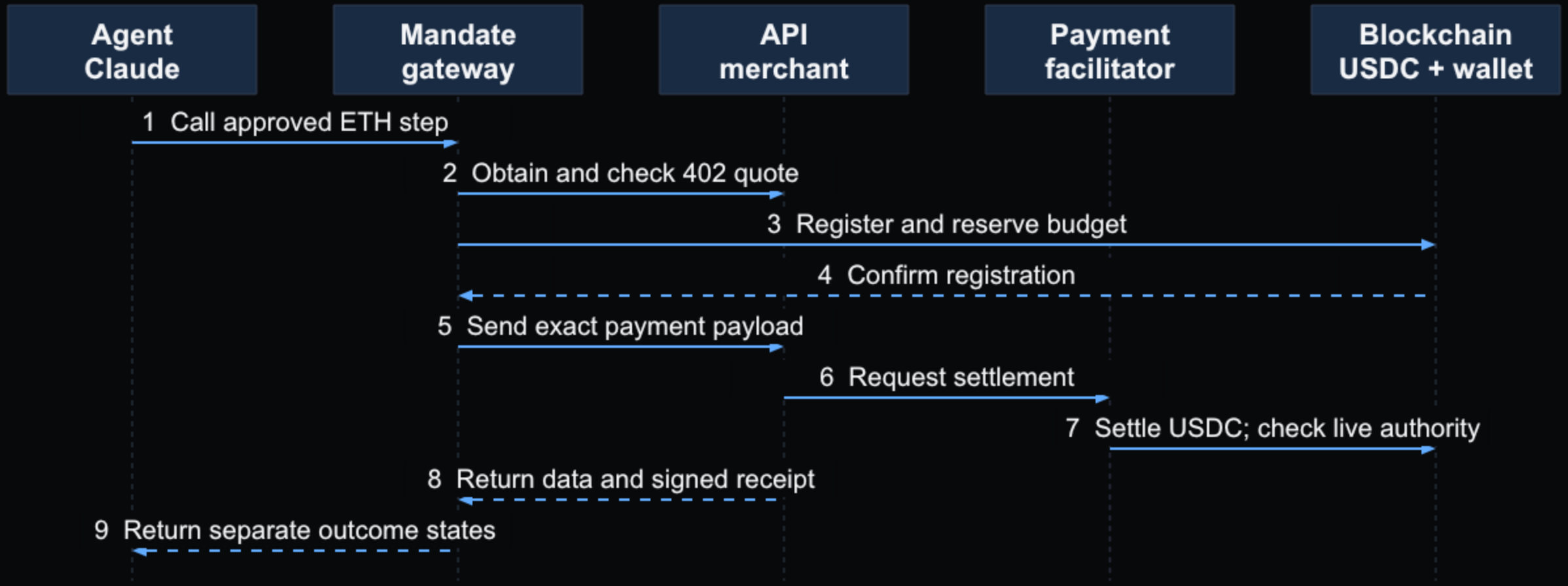
The BTC service was not bought and the session remains active until 23:27 SGT.

Fixed sample dataset. Purchases use test USDC.

Account preparation and delegation



Quote, registration and settlement



Contract-enforced financial scope

Check	Enforcement
Scope	Fixed asset, approved merchant and request
Shared budget	Atomic task and session reservations
Business retry	One used-step flag and deterministic nonce
Settlement	Registered digest with current version and epoch
Safe release	Expired payment with unused token nonce

One payment through a delivery failure

Observation	Payment	Service	Result
First paid response	paid	failed	unchecked
Same-order recovery	paid	returned	valid

0.01 test USDC

One settlement hash. One payment nonce.

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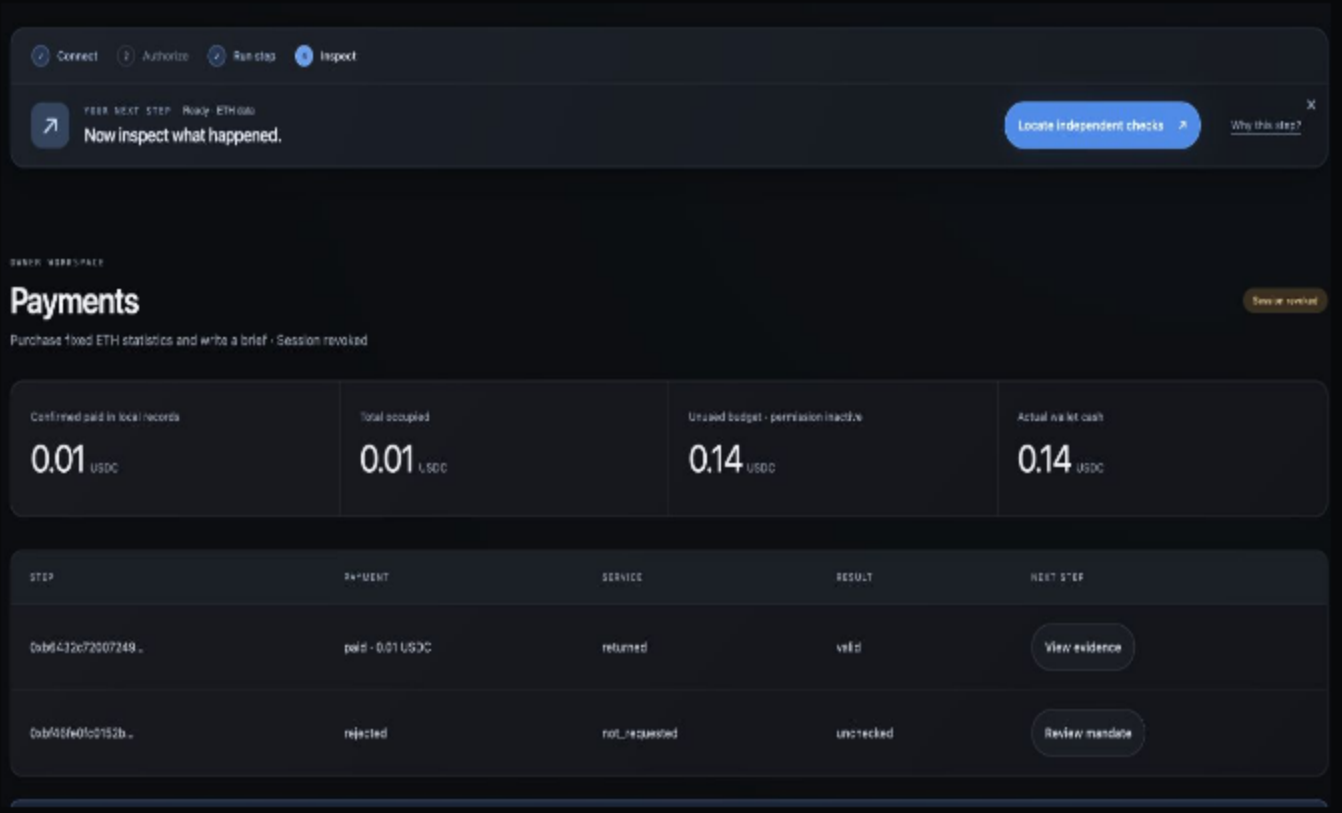
Revocation stops the next purchase

Session revoked

Unused BTC step rejected before registration

0 new transactions

0.14 test USDC remains



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Evidence guarantees and their limits

Confirmed in this run	Still unproven
Registration authority and budget	Independent HTTPS provenance
One matching USDC payment	Complete unobserved history
Signed offer, receipt and fixed result	Isolation from an administrator on the same host

Required financial checks passed. Overall report: incomplete.

Evaluation and measured overhead

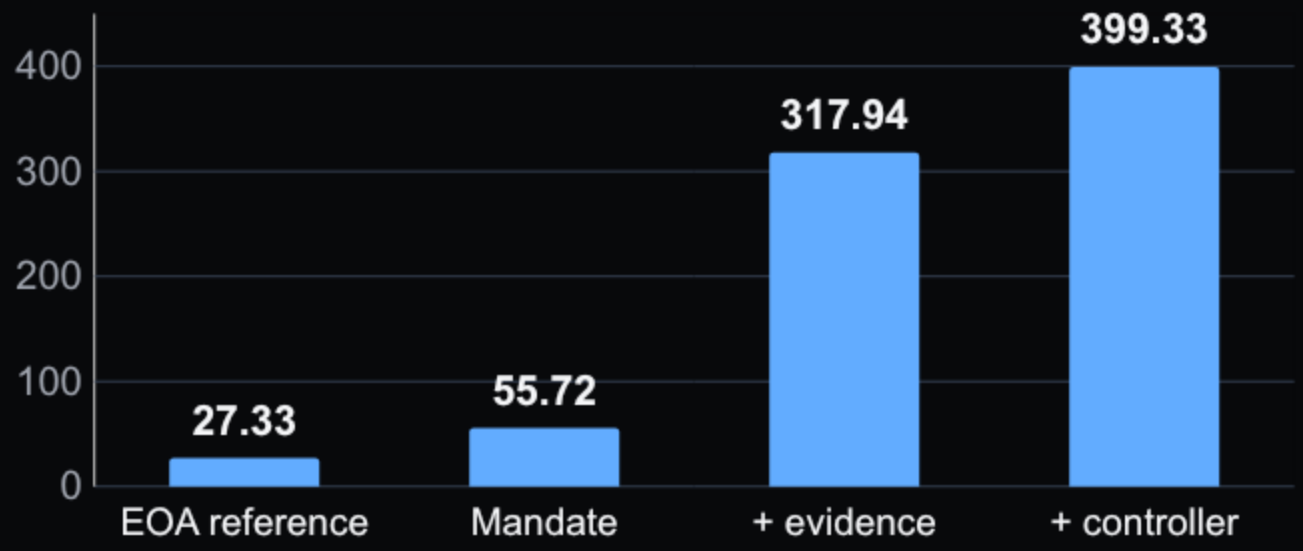
120 tasks

132 settlements; no duplicate payments

10,000 invalid calls

20 explicit invalid categories
6 legal controls

Local full-task median (ms)



n=30 per group on a local chain. Setup excluded; export/verification included where applicable.

Echo, Mandate and the research proposal

Foundation	Contribution to this project
Echo v2	Operation-bound signatures and controlled swap reference
Mandate protocol	Registered USDC payments, shared budgets and recovery
Research proposal	Linked evidence, selective disclosure and host-interference limits

The proposal motivates the research; its complete intended guarantees remain outside this delivery.

Current product and next priorities

Working today

Base Sepolia purchase through Claude Desktop, with owner control and recoverable records.

Next release priorities

Signed installation and updates, persistent local practice, broader service compatibility.

Testnet engineering release. Mainnet and an external human security audit remain pending.



Agent commerce on your terms

Owner-approved scope with reviewable payment outcomes